ZABIR Z

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SUMMARY

Results-driven Electronics and Communication Engineering student with a solid foundation in building IoT, Embedded Systems, VLSI Design, Computer Vision, and 3D/CAD Design. Experienced in architecting and deploying end-to-end hardware-software solutions using Arduino, ESP32, C/C++, Python, and industry-standard CAD tools — from intelligent automation systems to AI-powered healthcare monitoring devices. Backed by 4 professional internships and multiple impactful projects, with a track record of delivering technically complex, real-world solutions. Passionate about pushing the boundaries of embedded intelligence, spatial design, and smart systems to create measurable impact across the electronics and technology landscape.

TECHNICAL SKILLS

Core Domains : Internet of Things(IoT), Embedded System, PCB Design, VLSI Design, 3D CAD Design, Computer vision, Image processing

Languages : C, C++, Html, Python

Hardware : Arduino, Esp32, Sensors, I2C,Webcam

Tools : Arduino IDE, Proteus, VS Code, GitHub, AdobePhotoshop, openCV, Blynk, Numpy

Soft Skills : Communication & Teamwork, Time Management, Adaptability & Flexibility, Leadership Skills

EDUCATION

2023 - 2027 **B.E. – Electronics and Communication Engineering**
Gnanamani College of Technology | CGPA: 8.0 / 10

2022 - 2023 **H.S. – Maths, Computer Science**
TN State Board | Scored 75.5%

PROJECTS

Smart Home Automation System (IoT)

Stack: ESP32,Relay, Wi-Fi Module, Sensors

- Designed and developed an IoT-based smart home system using ESP32 and sensors. Enabled remote monitoring and control of home appliances.
- Implemented real-time automation logic to improve efficiency.

AI & IoT Based Healthcare Monitoring System

Stack: Esp32, IoT Sensors, Python

- Built an IoT-enabled health care monitoring system integrating sensors with AI-based data analysis. Monitored vital parameters such as pulse and temperature with real-time alerts.
- Enhanced health data tracking and intelligent decision support.

Distance Measurement Using Ultrasonic Sensor

Stack: ArduinoUno,HC-SR04 Sensor, LCD Display

- Developed a distance measurement system using ultrasonic sensor, Arduino, and LCD display.
- Built a real-time distance measurement system with ±1 cm accuracy over a 2–400 cm range. Displayed live readings on 16x2 LCD and calibrated sensor output for reliable environmental use.

AI-Based Soil Data Communication and Disaster Management

Stack: ESP32, IoT, Embedded C, Sensors, Rule-Based AI

- Developed an IoT-based soil monitoring system using ESP32 and sensors to collect real-time environmental data for agriculture and disaster prevention.
- Implemented rule-based AI for intelligent analysis of soil, rain, and water levels to detect risks like floods and landslides and send alerts.

INTERNSHIP EXPERIENCES

Jul 2026

Embedded Systems Intern
NSIC | Chennai

- Completed hands-on training in Embedded Systems using Arduino and microcontroller-based development. Gained practical experience in embedded C programming, sensor interfacing, and electronic circuit implementation

Jul 2026

COMPUTER VISION Intern
NSIC | Chennai

- Completed a one-month Computer Vision Internship at NSIC, Chennai, gaining hands-on experience in Python, OpenCV, image processing, object detection, and real-time computer vision application development.

Dec 2025 – Jan 2026

Embedded Systems Intern
UnitedSoftTech | Salem

- Worked on embedded system design and development, gaining hands-on experience in microcontrollers, sensors, and circuit interfacing.
- Implemented basic embedded programming concepts for real-time applications.

Oct 2025 – Dec 2056

Very Large Scale Integration - intern
Smarted Innovation | Bangalore | Online

- Gained practical exposure to VLSI fundamentals including digital logic and CMOS basics.
- Understood IC design flow such as specification, design, and basic verification.
- Strengthened core knowledge relevant to semiconductor and VLSI industries.

Jul 2025

Internet of Things - intern
Emglitz Technologies | Salem

- Learned and applied IoT fundamentals using Arduino and ESP32. Integrated sensors for real-time monitoring and control applications.
- Developed basic IoT mini-projects for practical implementation.

Jan 2025

Web Development - intern
Odugaa Tech | Salem

- Gained hands-on exposure to front-end fundamentals — HTML, CSS, and basic responsive layout design.

CERTIFICATIONS

PCB Design Workshop :
Sep 2024

Mastering PCB Design: A Hands-on Training Workshop” by IndiGuard Systems Pvt. Ltd. at Gnanamani College of Technology.

PROCOMPANION Workshop :
Sep 2024 | Sponsor Qualcomm

Attended a technical workshop at National Institute of Technology (NIT) on Procompanion, gaining exposure to advanced technical concepts and hands-on learning.

Workshop :
Sep 2024

Attended National Level Symposium on "Generative AI for Electronics and Communication" at KSR Engineering College, gaining insights into AI-driven innovations in the ECE domain.

Project Expo :
Mar 2026

Participated in i-SYM’26 (IEEE) Project Expo at Hindusthan College of Engineering and Technology, showcasing a technical project and enhancing presentation skills.

Completed online and hands-on learning programs related to technology and skill development. resented technical ideas through PPT presentations and actively engaged in inter-college workshops to improve communication, teamwork, and technical knowledge.

Achievements & Activities

Jan 2025

Paper Presentation :
Technical Symposium | 2nd prize

Won 2nd Prize in Paper Presentation on Haptic Technology at AVS Engineering College, demonstrating strong research and technical communication skills.

Sep 2025

SIH :
Government Hackathon

Participated in Smart India Hackathon (SIH) with an innovative IoT-based solution, collaborborating in a team to address real-world problems.